

Production Planning and Execution (PP)

This case study explains an integrated production planning and execution process in detail and thus fosters a thorough understanding of each process step and underlying SAP functionality.

Product

S/4HANA 2022
Global Bike

Fiori 3.0

Level

Undergraduate
Graduate
Beginner

Focus

Production Planning and
Execution

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Version

4.2

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MOTIVATION

The data entry requirements in the production planning exercises (PP 1 through PP 6) were minimized because much of the data already existed in the SAP system. This stored data, known as master data, simplifies the processing of business transactions. Examples for this, are material master data, bills of materials, and routings.

In this case study, we will create consumption values for a finished product to plan and process a complete manufacturing cycle.

PREREQUISITES

Before you use this case study, you should be familiar with navigation in the SAP system.

In order to successfully work through this case study, it is not necessary to have finished the PP exercises (PP 1 through PP 6). However, it is recommended.

NOTES

This case study uses the Global Bike data set.



Process Overview

Learning Objective Understand and perform a manufacturing process cycle.

Time 200 min

Scenario In order to experience a complete manufacturing process you will take on different roles within the Global Bike Group, e.g., production supervisor, shop floor worker and plant manager. Overall, you will be working in the Materials Management (MM) and the Production Planning and Execution (PP) departments.

Employees involved

- Jun Lee (Production Manager)
- Hiro Abe (Plant Manager)
- Lars Iseler (Shop Floor Worker 2)
- Susanne Castro (Goods Receipt Clerk)
- Sanjay Datar (Warehous Employee)
- Michael Brauer (Shop Floor Worker 4)
- Jamie Shamblin (Controller)

Before you can start forecasting demand for your touring bike product group, changes in the material master record of the bikes need to be maintained.

Afterwards you will create a 12-month sales and operations plan (SOP) for your product group, receive the production relevant goods from the warehouse storage location and issue them to the production order. To conclude the process, the production is confirmed as complete, the finished goods are received into the warehouse and costs assigned to the production order are analyzed.

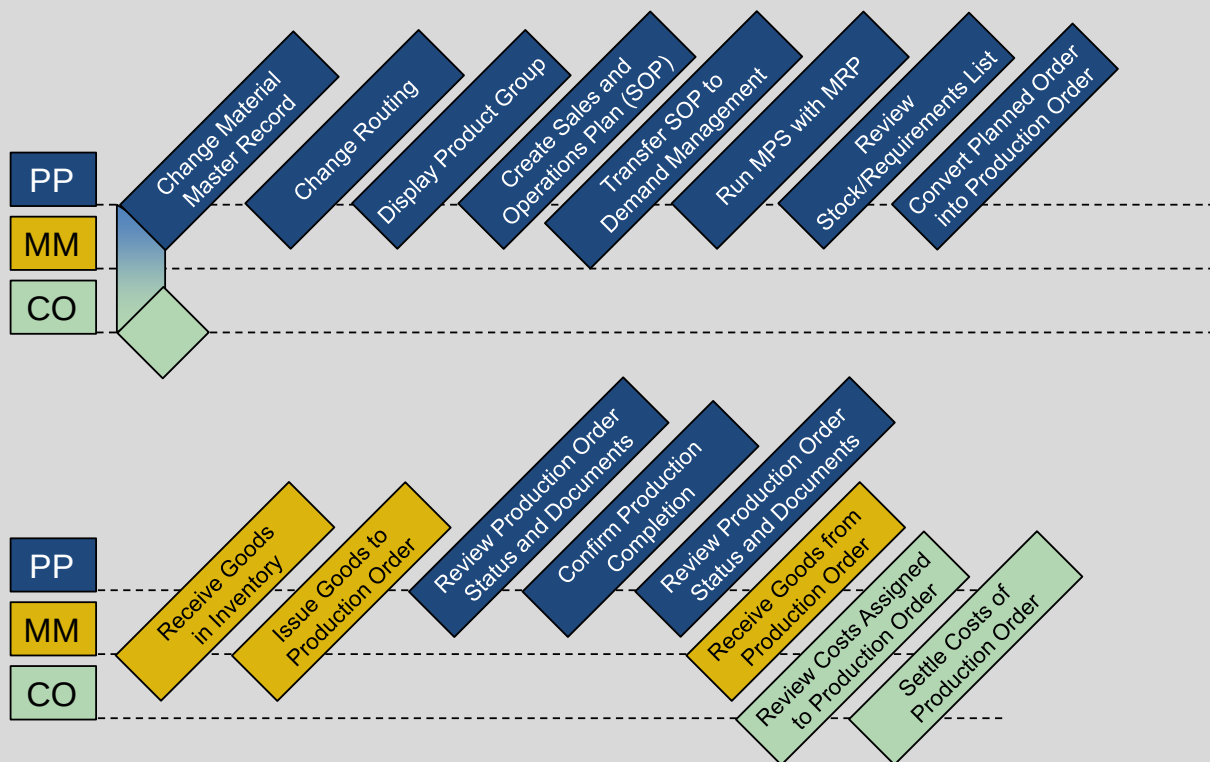


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Step 1: Change Material Master Record

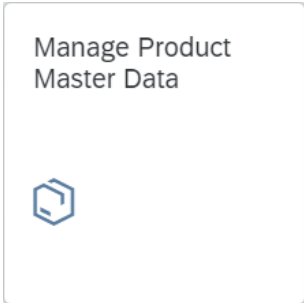
Task Prepare a material master record for Demand Planning. **Time** 20 min

Short Description In order to plan Global Bike’s deluxe touring bikes (black, silver and red) prepare their material master records by adding planning-relevant data to these records.

Name (Position) Jun Lee (Production Manager)

To change the views of a material, in the space *Production Planning and Execution* space and the role of *Production Manager* use the *Manage Product Master Data* app.

Manage Product Master Data



In the search screen, enter **DXTR*###** (replace ### with your three-digit number) in the search field.

DXTR*###

Manage Product Master Data

Standard

Editing Status: Product: Product Description: GTIN: Product Group:

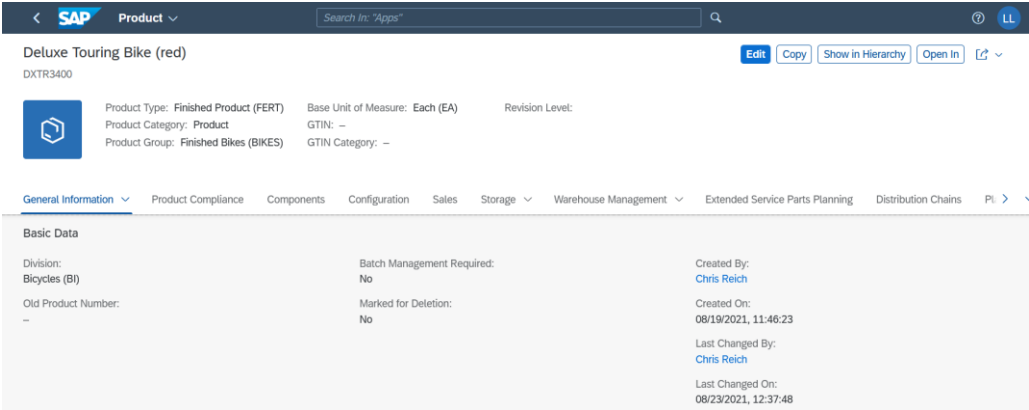
Product Category:

Adapt Filters (1)

Press . Your various Deluxe Touring Bikes will be displayed.

Products (3)					
	Description / ID	Group / Type	GTIN	Product Category	Last Changed
☐	Deluxe Touring Bike (black) DXTR1400	Finished Bikes (BIKES) Finished Product (FERT)		Product	08/23/2021, 12:36:25 Chris Reich
☐	Deluxe Touring Bike (silver) DXTR2400	Finished Bikes (BIKES) Finished Product (FERT)		Product	08/23/2021, 12:37:10 Chris Reich
☐	Deluxe Touring Bike (red) DXTR3400	Finished Bikes (BIKES) Finished Product (FERT)		Product	08/23/2021, 12:37:48 Chris Reich

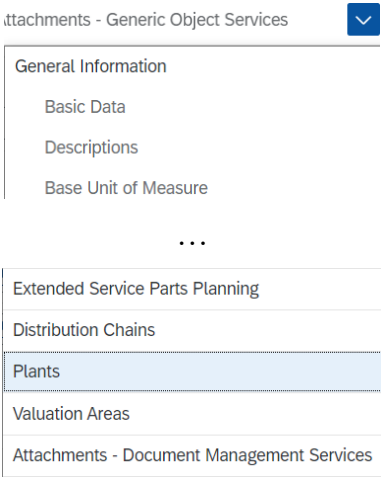
Click the line of **Deluxe Touring Bike (red)** (DXTR3##) to open the details of the product.



Press **Edit** to switch to the edit mode.

Use the pull-down menu  to select the *Plants* section. The window automatically scrolls to the correct position.

Plants



You will see a list of all plants for which the product has been defined.

Plants				
Plants (5)				
Plant	MRP Type	MRP Contr...	Marked for ...	
☐ Plant Dallas (DL00)	M1 	000 	☐	>
☐ Plant Heidelberg (HD00)	M1 	000 	☐	>
☐ DC Hamburg (HH00)	M1 	000 	☐	>
☐ DC Miami (MI00)	M1 	000 	☐	>
☐ DC San Diego (SD00)	M1 	000 	☐	>

Press  at the end of the line with the DL00 plant to open the plant-specific product master data.

SAP

Plant

Deluxe Touring Bike (red) /

Plant Dallas

DL00

Product Type: Finished Product (FERT)

Product Category: Product

Product Group: Finished Bikes (BIKES)

Base Unit of Measure: Each (EA)

GTIN: --

GTIN Category: --

Reference Product: --

Fiscal Year of Current Period: 2023

Current Period: 8

General Information

International Trade

Sales

Quality Management

MRP Data

Advanced Planning

Extended Service Parts Planning

Forecasting

General Data

Plant: Plant Dallas (DL00)

Batch Management Required: No

Marked for Deletion: ☐

Plant-Specific Product Status:

Serial Number Profile:

Min. Temperature:

0.00

Valid From Date for Status:

MMkkJJyyyy

Negative Stocks Allowed in Plant: ☐

Max. Temperature:

0.00

Profit Center:

Stock Determination Group:

Unit of Issue:

Is Excise Tax Relevant: ☐

Select the *MRP Data* area. The window automatically scrolls to the correct position. Enter **40** in the *Strategy Group* field (Planning with final assembly).

MRP Data
40

MRP Data

MRP Type: * MPS, fixing type -1- (M1)

ABC Indicator:

Reorder Point:

0

 EA

MRP Controller: DL MRP Controller (000)

MRP Group:

Planning Cycle:

Availability Check: Individ. Requirement (02)

Strategy Group: Planning with final assembly (40)

Planning Time Fence:

7

 DAY

Select the *Forecasting* → *Required Periods* area. If the tab is not visible, you can use the pull-down menu again.

Forecasting →
Required Periods

SAP

Plant

Deluxe Touring Bike (red) /

Plant Dallas

DL00

Advanced Planning

Extended Service Parts Planning

Forecasting

Forecasting

Forecasting

Required Periods

Control Data

In the *Periods for Initialization* field enter **12**. Scroll down to the next area *Control data*.

12

In the *Control Data* area below, click on the value help icon of the *Optimization Level* field and select the optimization level **F** - Fine (high optimization level). Then select **Parameter Optimization**.

F
Parameter
Optimization

Now assign the smoothing factors. Enter **0.20** for the *Alpha Factor* (base value), **0.10** for the *Beta Factor* (trend value), **0.30** for the *Gamma Factor* (seasonal index) and **0.30** for the *Delta Factor* (MAD).

0.20
0.10
0.30
0.30

Required Periods

Historical Periods:

120

MON

Forecast Periods:

12

MON

Periods per Seasonal Cycle:

12

MON

Periods for Initialization:

12

MON

Fixed Periods:

0

MON

Control Data

Initialization Indicator:

Initialization by system (X)

Model Selection Indicator:

Optimization Level:

Fine (high optimization level) (F)

Tracking Limit:

4.000

Model Selection Procedure:

Analytical model selection procedure (2)

Weighting Group:

Reset Forecast Model Automatically:

Parameter Optimization:

☒

Correction Factors:

☐

Alpha Factor:

0.20

Beta Factor:

0.10

Gamma Factor:

0.30

Delta Factor:

0.30

Select **Apply** to save the plant-specific data for plant DL00.

Click **Save** to save your changes to the red Deluxe Touring Bike.

The SAP system updates the master data record for material DXTR3### and displays a corresponding message.

Master data record saved.

Select **<** to return to the Manage Product Master Data screen.

Repeat the same procedure for the silver and black Deluxe Touring Bike. Start with the silver one (**DXTR2###**) and then change the black bike (**DXTR1###**).

Click **SAP** to return to the SAP Fiori Launchpad.

DXTR2###
DXTR1###

Step 2: Change Routing

Task Change a routing for a finished good.

Time 15 min

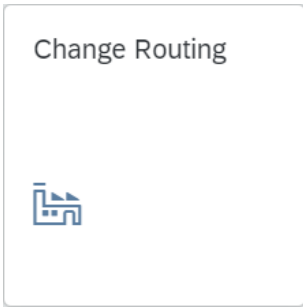
Short Description Change the routing for your red Deluxe Touring bike.

Name (Position) Jun Lee (Production Manager)

After the operational steps are laid out, the components must be allocated to the individual operations. This is a progressive process where each operation builds off the materials that entered production in the previous operations.

To change the routing, in the space *Production Planning and Execution* space and the role of *Production Manager* use the *Change Routing* app.

Change Routing



Enter the material number of your red Deluxe Touring bike (**DXTR3###**). In the *Plant* field, enter the Global Bike plant number in Dallas (**DL00**). Also, please make sure that the field *Group* is empty.

DXTR3###
DL00

< SAP Change Routing: Initial Screen

Menu ▾

Material: DXTR3400

Plant: DL00

Sales Document: Sales Document Item:

WBS Element:

Group:

Then press Operations.

< SAP

Change Routing: Operation Overview

Search In: "Apps"

Menu

<

>

Reference

Work center

Allocation

Detail

Sequences

PRT

Group: 7204

Group Counter: 1

Material: DXTR3400

Deluxe Touring Bike (red)

Sequence: 0

Sequence Desc.:

Operation Overview

Op...	SOp	Work Ce...	Plant	* C...	Standard...	Description	Lo...	PRT	Cl...
☐	0010	ASSY1000	DL00	ASSY		Material staging	☐	☐	☐
☐	0020	ASSY1000	DL00	ASSY		Attach seat to frame	☐	☐	☐
☐	0030	ASSY1000	DL00	ASSY		Attach handle bar assembly	☐	☐	☐
☐	0040	ASSY1000	DL00	ASSY		Attach derailleur gear assm. to wheel	☐	☐	☐
☐	0050	ASSY1000	DL00	ASSY		Attach front and rear wheels to chain	☐	☐	☐
☐	0060	ASSY1000	DL00	ASSY		Attach brakes	☐	☐	☐
☐	0070	ASSY1000	DL00	ASSY		Attach peddles	☐	☐	☐
☐	0080	INSP1000	DL00	ASSY		Test bike	☐	☐	☐
☐	0090	PACK1000	DL00	ASSY		Disassemble	☐	☐	☐
☐	0100	PACK1000	DL00	ASSY		Pack bike	☐	☐	☐
☐	0110	PACK1000	DL00	ASSY		Move to storage	☐	☐	☐

Note A routing is defined by the routing group and the routing group counter. Moreover, the routing contains a reference to the material whose production is described by the routing.

Besides the standard sequence, it can also have parallel or alternative sequences. Alongside the standard values, the routing also contains time elements that are relevant for scheduling operations. Each operation in the routing may contain its own base quantity, to which these time elements may refer.

Select **Allocation** to display a list of all components. If this is not displayed, you will find the entry in the pull-down menu under **Menu ▶ Goto ▶ Allocation**.

Select the rows Touring Frame-Red (**TRFR3##**) and Touring Seat Kit (**TRSK1##**).

TRFR3###
TRSK1###

Item Overview											
	Pl...	Le...	Path	It...	Component	Quantity	Sort String	Un...	It...	Ba...	Activity Seq.
☐		0	0	0010	TRWA1400	2		EA	L		Touring Aluminum Wheel Assembly
☒		0	0	0020	TRFR3400	1		EA	L		Touring Frame-Red
☐		0	0	0030	DGAM1400	1		EA	L		Derailleur Gear Assembly
☒		0	0	0040	TRSK1400	1		EA	L		Touring Seat Kit
☐		0	0	0050	TRHB1400	1		EA	L		Touring Handle Bar
☐		0	0	0060	PEDL1400	1		EA	L		Pedal Assembly
☐		0	0	0070	CHAN1400	1		EA	L		Chain
☐		0	0	0080	BRKT1400	1		EA	L		Brake Kit
☐		0	0	0090	WDOC1400	1		EA	L		Warranty Document
☐		0	0	0100	PCKG1400	1		EA	L		Packaging

Press **New Assignment**. In the popup that appears, enter **0020** for *Activity* and confirm the entry with ☒.

0020

New Assignment

Assign to

Activity: 0020Sequence: 0

✓

Oper./act. list

🔍

✗

Back in the *Routing Change: Material Component Overview* you can see that both components have now been assigned to *Activity 0020*.

Item Overview

	Ph...	Le...	Path	Ite...	Component	Quantity	Sort String	Un...	It...	Ba...	Activity	Seq.
☐	☐	0	0	0010	TRWA1400	2		EA	L			
☐	☐	0	0	0020	TRFR3400	1		EA	L	☐	0020	0
☐	☐	0	0	0030	DGAM1400	1		EA	L			
☐	☐	0	0	0040	TRSK1400	1		EA	L	☐	0020	0
☐	☐	0	0	0050	TRHB1400	1		EA	L			
☐	☐	0	0	0060	PEDL1400	1		EA	L			
☐	☐	0	0	0070	CHAN1400	1		EA	L			
☐	☐	0	0	0080	BRKT1400	1		EA	L			
☐	☐	0	0	0090	WDOC1400	1		EA	L			
☐	☐	0	0	0100	PCKG1400	1		EA	L			

Repeat this process for all other components and assign them to the operations below.

Component	Operation
TRHB1### (touring handlebar)	0030
TRWA1### (touring aluminum wheel assembly)	0040
DGAM1### (derailleur gear assembly)	0040
CHAN1### (chain)	0050
BRKT1### (brake kit)	0060
PEDL1### (pedal assembly)	0070
WDOC1### (warranty document)	0100
PCKG1### (packaging)	0100

- TRHB1### - 0030
- TRWA1### - 0040
- DGAM1### - 0040
- CHAN1### - 0050
- BRKT1### - 0060
- PEDL1### - 0070
- WDOC1### - 0100
- PCKG1### - 0100

Item Overview

	Ph...	Le...	Path	Ite...	Component	Quantity	Sort String	Un...	It...	Ba...	Activity	Seq.
☐	☐	0	0	0010	TRWA1400	2		EA	L	☐	0040	0
☐	☐	0	0	0020	TRFR3400	1		EA	L	☐	0020	0
☐	☐	0	0	0030	DGAM1400	1		EA	L	☐	0040	0
☐	☐	0	0	0040	TRSK1400	1		EA	L	☐	0020	0
☐	☐	0	0	0050	TRHB1400	1		EA	L	☐	0030	0
☐	☐	0	0	0060	PEDL1400	1		EA	L	☐	0070	0
☐	☐	0	0	0070	CHAN1400	1		EA	L	☐	0050	0
☐	☐	0	0	0080	BRKT1400	1		EA	L	☐	0060	0
☐	☐	0	0	0090	WDOC1400	1		EA	L	☐	0100	0
☐	☐	0	0	0100	PCKG1400	1		EA	L	☐	0100	0

Apply your changes with

Save

. The system issues a message that the routing has been saved.

✔ Routing was saved with group 7204 and material DXTR3400.

Click  to return to the SAP Fiori Launchpad.



Step 3: Display Product Group

Task Display a product group.

Time 5 min

Short Description Display the product group (product family) for all your Deluxe Touring bikes.

Name (Position) Jun Lee (Production Manager)

A product group (product family) supports high-level planning. This way, it is not necessary to delve into the minutia of creating planning forecasts for every material in the company.

Product group

To view the Deluxe Touring bike product group, in the space *Production Planning and Execution* space and the role of *Production Manager* use the *Display Product Group* app.

Display Product Group



In the *Display Product Group: Initial Screen*, in the *Product group* field find and select your group for deluxe touring bikes. In order to do so, press the search icon  (or pressed F4), enter **###*** in the *Material description* field. Remember to replace **###** with your three-digit number, e.g., enter 009* if your number is 009. Enter **DL00** as *Plant*.

###*

DL00

Name of the product group (1)

Search and Select

D: Find product group via MRP controller ▾

Go

Hide Filters

MRP Controller:

Material description:

Language Key:

Product group:

Plant:

400*

EN

DL00

Items (0)

Then, press Enter or click on  to display the search results.

Name of the product group (1)

Search and Select

D: Find product group via MRP controller ▾

Go Hide Filters ⚙️ ▾

MRP Controller:

Material description:

Language Key:

Product group:

Plant:

Items (5)

Find Find next Add to Favorites ⚙️ ▾

MRP Contr...	Material description	Langu...	Product gr...	P...
☒	400 PRODUCT GROUP BICYCLES	EN	PG-BIKE400	DL00
☐	400 PRODUCT GROUP DELUXE TOURING BICYCLE	EN	PG-DXTR400	DL00
☐	400 PRODUCT GROUP OFFROAD BICYCLES	EN	PG-ORBK400	DL00
☐	400 PRODUCT GROUP PROFESSIONAL TOURING	EN	PG-PRTR400	DL00
☐	400 PRODUCT GROUP TOURING	EN	PG-TRBK400	DL00

You will see a list of all your product groups, e.g., for mountain bikes or touring bikes. Select the group of Deluxe Touring Bikes (**PG-DXTR##**). Then click **OK** to apply the selection.

PG-DXTR###

Now that the correct *Product group* (**PG-DXTR###**) is filled in, make sure that *Plant* **DL00** is entered.

DL00

<  Display Product Group: Initial Screen

Menu ▾

Product group:

Plant:

Then, press Enter to display the product group details.

On this screen, you can see that this product group defines proportions for three different bikes: the black, silver and red deluxe touring bike. For the black bike, a share of 40% will be considered and 30% for the silver and the red bikes each.

<  Display Product Group: Members (Materials)

Menu ▾

Product group: PG-DXTR400

Plant: DL00: Plant Dallas

Base Unit: EA

Member number	Plnt	Unit conv.	Aggr.fact.	Proportion	UoM	V	M	Fx	Short Text	MTyp
DXTR1400	DL00	1	1	40	EA			☐	Deluxe Touring Bike (black)	FERT
DXTR2400	DL00	1	1	30	EA			☐	Deluxe Touring Bike (silver)	FERT
DXTR3400	DL00	1	1	30	EA			☐	Deluxe Touring Bike (red)	FERT

Click  to return to the SAP Fiori Launchpad.



Step 4: Create Sales and Operations Plan (SOP)

Task Create a sales and operations plan for a product group.

Time 20 min

Short Description Create a 12-month sales and operations plan (SOP) for your product group.

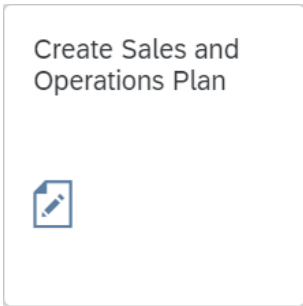
Name (Position) Jun Lee (Production Manager)

A sales and operations plan (SOP) is a planning tool that is used to consolidate data for forecasting future sales and production levels as well as the methods needed to meet those requirements. In this task, our SOP will be based on historical consumption values taken from a fixed period. This is in contrast to forecasting within a real-life system which would base the prediction on previous periods and their respective consumption.

Sales and operations plan

To create an SOP, in the space *Production Planning and Execution* space and the role of *Production Manager* use the *Create Sales and Operations Plan* app.

Create Sales and Operations Plan



Make sure that *Product group* **PG-DXTR###** and *Plant* **DL00** are entered. Then, select **Active version**.

PG-DXTR###
DL00

Record the version number:_____

In the menu bar, select:

Menu ► Edit ► Create sales plan ► Forecast...

Select **Period intervals**, *Forecast* from **current period/current year** to **previous period/next year**, *Historic Data* from **06/2017** to **03/2021**, *Forecast execution* **Aut. model selection**. Compare your screen with the one below before clicking on **Historical...** to view the historical values.

Period intervals
current period/current
year
previous period/next
year
04/2017
03/2021
Aut. model selection

Forecast: Model Selection

Periods

☒ Period intervals

Forecast

* Fr... : 08/2023

* To: 07/2024

Historical data

* Fr... : 06/2017

* To: 03/2021

☐ No. of periods

No. of forecast periods: 0

No. of historical values: 60

Forecast execution

☐ Constant models

☐ Seasonal models

☐ Trend models

☐ Season. Trend Models

☒ Aut. model selection

☐ Historical

Forecast parameters

Profile: SAP

Forecasting

Historical...

Forecast profile...

Version...

You will get an overview of the passed periods in the specified time range.

Forecast: Historical Values

Historical values

Period	Val. fld	Corr.value	F	C
M 03/2021	333	333	☐	☐
M 02/2021	340	340	☐	☐
M 01/2021	363	363	☐	☐
M 12/2020	310	310	☐	☐
M 11/2020	276	276	☐	☐
M 10/2020	283	283	☐	☐
M 09/2020	306	306	☐	☐
M 08/2020	283	283	☐	☐

Forecasting

Correct

Click on

Forecasting

.

In the next Popup the system selected *Trend and season*. Click on

Forecasting

 again.

In the next pop-up you can see that the system tested and found Seasonal and Trend tendencies in the past consumption data and has applied a Seasonal Trend Model.

Forecast: Results

Basic value: 320.895

Trend value: 6

MAD: 17

Error total: 73

Forecast results

Period	Orig. HV	Corr. HV	Ex-post FV	Orig. FV	Corr. FV	Season	F	C
M 04/2021				318	318	0.97	☐	☐
M 05/2021				343	343	1.03	☐	☐
M 06/2021				299	299	0.89	☐	☐
M 07/2021				314	314	0.91	☐	☐
M 08/2021				348	348	1.00	☐	☐
M 09/2021				382	382	1.08	☐	☐
M 10/2021				352	352	0.98	☐	☐

Time series characteristic changed

✓

📄

🔄 Forecasting

⚙️ User exit

Press , the sales forecast has been transferred to the SOP. Look at the planning table. Note that your values may differ from the screenshot.

<

SAP

Change Rough-Cut Plan

Search In: "Apps"

Menu

Characteristic

Product group: PG-DXTR400

400 Product Group Deluxe Touring Bicycle

Plant: DL00

Version: A00 Active version

Active

SOP: plan individual product group

Planning Table	Un	M 08/2023	M 09/2023	M 10/2023	M 11/2023	M 12/2023	M 01/2024	M 02/2024	M 03/2024
Sales	EA	483	527	484	471	534	582	536	521
Production	EA								
Stock level	EA	-483	-1010	-1494	-1965	-2499	-3081	-3618	-4140
Target stock level	EA								
Range of Coverage									
Target days' supply									

As *Target day's supply* enter 5 for each forecasted period (for 12 periods).

SOP: plan individual product group

Planning Table	Un	M 08/2023	M 09/2023	M 10/2023	M 11/2023	M 12/2023	M 01/2024	M 02/2024
Sales	EA	483	527	484	471	534	582	536
Production	EA							
Stock level	EA	-483	-1010	-1494	-1965	-2499	-3081	-3618
Target stock level	EA							
Range of Coverage								
Target days' supply		5	5	5	5	5	5	5

In a production plan, you plan the quantities you need to produce in order to meet your sales plan. The system then calculates stock levels and days' supply for each period on the basis of the sales and production quantities and any

target data. There are several different planning strategies available, which differ in the production values and the stock levels proposed.

As the SOP is a high-level planning, discrete production values are not necessary. The SAP system calculates discrete numbers once the SOP is transferred to the Demand Management.

In the menu bar, select:

Menu ► Edit ► Create production plan ► Synchronous to sales

Note the change in the *Production* and in the *Stock* level rows. The production plan is created to match the sales forecast.

SOP: plan individual product group

Planning Table	Un	M 08/2023	M 09/2023	M 10/2023	M 11/2023	M 12/2023	M 01/2024	M 02/2024
☐ Sales	EA	483	527	484	471	534	582	536
☐ Production	EA	483	527	484	471	534	582	536
☐ Stock level	EA							
☐ Target stock level	EA							
☐ Range of Coverage								
☐ Target days' supply		5	5	5	5	5	5	5

In the system menu, select

Menu ► Edit ► Create production plan ► Target day's supply

Note the impact on the production plan and stock levels. Production levels are generated to match the sales plus the amount needed to be put into stock to meet the target days' supply specifications.

Review the Planning Table (your numbers may be different).

SOP: plan individual product group

Planning Table	Un	M 08/2023	M 09/2023	M 10/2023	M 11/2023	M 12/2023	M 01/2024	M 02/2024
☐ Sales	EA	483	527	484	471	534	582	536
☐ Production	EA	560	536	474	471	541	589	534
☐ Stock level	EA	77	87	78	78	86	93	92
☐ Target stock level	EA							
☐ Range of Coverage		5	5	5	5	5	5	5
☐ Target days' supply		5	5	5	5	5	5	5

Note Although the screen displays integer production values, the SAP system calculates with decimal precision. You can display the decimal places of a series by pressing F8. Then create the production plan.

Accept the SOP with . A system message appears and you return to the initial screen.

 Plan saved under version number A00

Click  to return to the SAP Fiori Launchpad.



Step 5: Transfer SOP to Demand Management

Task Transfer SOP to Demand Management.

Time 10 min

Short Description Transfer the Sales and Operations Plan to Demand Management.

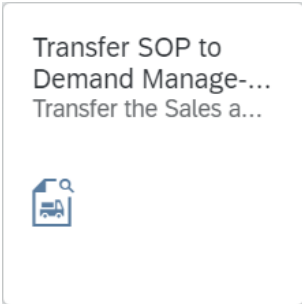
Name (Position) Jun Lee (Production Manager)

Demand Management is the tool used to disaggregate planning data from high-level plans down to the detailed planning level. For this task, planning for the Deluxe Touring Product Group will be broken down into the individual components that belong to this group.

Demand Management

To transfer sales/rough planning to Demand Management, in the space *Production Planning and Execution* space and the role of *Production Manager* use the *Transfer Planning Data to Demand Management* app.

Transfer Planning Data to Demand Management



Enter *Product group* **PG-DXTR###**, *Plant* **DL00**, and the version saved in the previous task (**A00**).

PG-DXTR###
DL00
A00

Select **Prod.plan for mat. or PG members as proportion of PG** and **Active**. Then, deselect the **Invisible transfer** indicator to present the disaggregation results on another screen allowing the planner to modify the results before saving them manually to Demand Management.

Prod.plan for mat. or PG members as prop. of PG
Active
~~Invisible transfer~~

<  Transfer Planning Data to Demand Management

Search In:

Menu ▾

Transfer now

* Product group:

PG-DXTR400

* Plant:

DL00

Version:

A00

Transfer strategy and period

☐ Sales plan for material or PG members

☐ Sales plan for mat. or PG members as proportion of PG

☐ Production plan for material or PG members

☒ Prod.plan for mat. or PG members as proportion of PG

From:

08/02/2023

To:

☐ Invisible transfer

Independent requirement specifications

Requirements type:

Version:

☒ Active

Select [Transfer now](#) and examine the Planned Independent Requirements generated for **DXTR1###**.

<  Plnd Ind. Reqmts: Planning Table

Search In: "Apps"

Menu ▾

Planning start: 08/02/2023

Planning End:

Table

Items

Schedule Lines

<< < > >>

	Material	MRP ...	V	A	BU	Req. Segment	M 08/2023	M 09/2023	M 10/2023	M 11/2023	M 12/2023
☐	DXTR1400	DL00	AG	☒	EA		224	214	190	188	216

Then click on [Save](#) to save the demand for the DXTR1###.

After saving, the system jumps to the independent requirement of the next material (DXTR2###). Now examine the independent requirement generated for DXTR2###.

Table

Items

Schedule Lines

<< < > >>

	Material	MRP ...	V	A	BU	Req. Segment	M 08/2023	M 09/2023	M 10/2023	M 11/2023	M 12/2023
☐	DXTR2400	DL00	AG	☒	EA		168	161	142	141	162

Continue with [Save](#). Finally, examine the planned independent requirement of material DXTR3###.

Also save this requirement with .

Note DXTR1### makes up 40%, DXTR2### makes up 30% and DXTR3### another 30% of the production plan created in your Sales and Operations Plan.

You will automatically return to the initial screen. The system also provides you with a message that the requirement has been saved. You can also see this in the other materials in advance.

 Requirement saved

Click  to return to the SAP Fiori Launchpad.



Step 6: Run MPS with MRP

Task Run Master Production Scheduling (MPS).

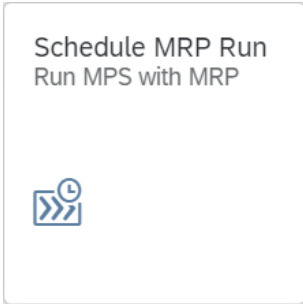
Time 10 min

Short Description Run Master Production Scheduling (MPS) to generate a series of planned orders that satisfy the requirements from SOP and demand management. Concurrently with MPS, the MRP materials will be processed leading to the generation of planned orders for dependent requirements that have been created by the BOM explosion process.

Name (Position) Jun Lee (Production Manager)

To start Master Production Scheduling, in the space *Production Planning and Execution* space and the role of *Plant Manager* use the *Schedule MRP Run - Run MPS with MRP* app.

Schedule MRP Run



Enter your *Material* **DXTR3###**, and as *Plant* **DL00**.

DXTR3###
DL00

The control parameters can be adopted and should be filled in by the system as follows:

- *Processing Key:* **NETCH** (Net-Change in Total Horizon)
- *Create Purchase Req.:* **2** (Purchase requisition in opening period)
- *SA Deliv. Sched. Lines:* **3** (Schedule lines)
- *Create MRP List:* **1** (MRP list)
- *Planning mode:* **1** (Adapt planning data (normal mode))
- *Scheduling:* **1** (Determination of Basic Date for Planned)

NETCH
2
3
1
1
1

Then, select **Display material list**.

Display material list

< **SAP** Single-Item, Multi-Level
Search

Menu v

* Material:

MRP Area:

Plant:

Scope of Planning

☐ Product group

MRP Control Parameters

* Processing Key: NETCH	Net Change in Total Horizon
* Create Purchase Req.: 2	Purchase requisitions in opening period
* SA Deliv. Sched. Lines: 3	Schedule lines
* Create MRP List: 1	MRP list
* Planning mode: 1	Adapt planning data (normal mode)
* Scheduling: 1	Determination of Basic Dates for Planned

Process Control Parameters

☐ Also Plan Unchanged Components

☐ Display Results Prior to Saving

☒ Display material list

☐ Simulation mode

Press Enter. A warning message will appear asking you to check input parameters. Press Enter to confirm and bypass the warning message.

Note In MRP, a net requirements calculation is executed in the planning run to determine whether a material shortage exists for a certain material. In addition, stock and fixed receipts that currently exist (for example, purchase orders, production orders, fixed purchase requisitions and planned orders) are compared with the safety stock and requirements. The result of this comparison is the quantity available for planning.

If the quantity available for planning is lower than zero, a material shortage exists. MRP reacts to material shortages by creating new procurement proposals (purchase requisitions and planned orders). The suggested procurement quantity results from the lot-sizing procedure that is set in the material master.

As soon as the planning run is completed, a result overview is displayed. Check the planning details of the result overview.

Menu   Materials

Statistics	
Materials planned	17
Materials with New Exceptions	17
Materials with Termination MRP List	

Parameters	
MRP Area	DL00
Plnt	DL00
Processing Key	NETCH
Create Purchase Requisition	2
SA Schedule Line	3
Create MRP List	1
Planning Mode	1
Scheduling	1

Scroll down further, there you will see a detailed listing of all materials considered.

Ranking List of Materials with Highest CPU Times (in ms)						
Material			MRP Area		Plnt	
Runtime	Read	Net Calc.	BOM	LdTmeSched	Update	
DXTR3400			DL00		DL00	
5,689	403	935	532	0	597	
BRKT1400			DL00		DL00	
566	3	482	0	0	33	
TRWA1400			DL00		DL00	
376	3	3	281	0	88	
BOLT1400			DL00		DL00	
185	3	24	0	0	157	
CHAN1400			DL00		DL00	
105	3	71	0	0	31	
TRFR3400			DL00		DL00	
75	3	40	0	0	32	

Click  to return to the SAP Fiori Launchpad.



Step 7: Review Stock/Requirements List

Task Review the Stock/Requirements List.

Time 10 min

Short Description Review the Stock/Requirements List for your deluxe touring bike.

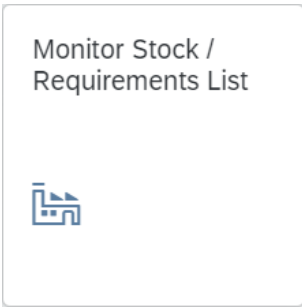
Name (Position) Lars Iseler (Shop Floor Worker 2)

The Stock/Requirements List is a list which dynamically changes whenever a transaction occurs using the given material. Display and review the Stock/Requirements List for all materials of the red deluxe touring bike on hand and the demand that exists against these products. The report shows that there is no stock and therefore nothing is available for use at this time.

Stock/Requirements List

To display the stock/requirements list, in the space *Production Planning and Execution* space and the role of *Shop Floor Worker* use the *Monitor Stock/Requirements List* app.

Monitor
Stock/Requirements List



On the *Individual access* tab, enter *Material DXTR3###* and *Plant DL00*. Click on **Continue** to display the associated stock/requirements list.

DXTR3###
DL00

< SAP

Stock/Requirements List: Initial Screen

Menu ▾

Individual access

Collective access

* Material: DXTR3400

Description: Deluxe Touring Bike (red)

MRP Area:

Plant: DL00 Plant Dallas

With filter: ☐

Currently, the system lists all entries as single rows.

Choose  (Switch to Period Totals). This will allow you to see the planned independent requirements, planned receipts, and ATP quantities based on time - days, weeks, or months.

Select to go back to the individual rows.

To view the details of the first planned order (PldOrd), select (Element Details).

GR pr.time: 0

Proc. Type:

Order Type: LA

 -> Prod.Ord. -> PartConvProdOrder -> Proc.Ord. -> SubProcOrd -> Pur Req. ★ ✖

Material	DXR3400	Deluxe Touring Bike (red)
MRP Area	DL00	Plant Dallas
Plant	DL00	Plant Dallas
Receipt Date	08/09/2023	
PO Quantity	168 EA	
Quantity Without Source	0 EA	

Planned dates	Material	Material Description	Material Memo	MRP Area	El	MRP element data	Rec./reqd quantity	Input Quantity	Unit
08/01/2023	DXTR3400	Deluxe Touring Bike (red)		DL00	PP	VSF	168	168	EA



Step 8: Convert Planned Order into Production Order

Task Convert a planned order into a production order.

Short Description Convert a planned order generated in the MPS/MRP run to a production order. The stock requirements list displays the suggested planned orders from the MPS run.

Name (Position) Lars Iseler (Shop Floor Worker 2)

Time 10 min

To convert planned orders into production orders, in the space *Production Planning and Execution* space and the role of *Shop Floor Worker* use the SAP Fiori app *Monitor Stock / Requirements List*.

Monitor Stock / Requirements List

Enter Material **DXTR3###**, Plant **DL00**, and click on **Continue**.

DXTR3###
DL00

Monitor Stock / Requirements List



Enter Material **DXTR3400**, Plant **DL00**, and click on **Continue**.

SAP Stock/Requirements List as of 13:47 hrs

Menu Show Overview Tree

Material: DXTR3400
Description: Deluxe Touring Bike (red)
MRP Area: DL00 Plant Dallas Ex. manuf.:
Plant: DL00 MRP type: M1 Material Type: FERT Unit: EA

GR ST On On Vendor Cust.

A...	Date	MRP e...	MRP element data	Rescheduling...	E...	Receipt/Reqmt	Available Qty	Pro...
08/02/2023	Stock						0	
08/01/2023	IndReq		VSF			168-	168-	
08/09/2023	----->		End of Planning Tim...					
08/09/2023	PldOrd		000000423/STCK	08/01/2023	30	168	0	0001
09/01/2023	PldOrd		000000424/STCK			161	161	0001
09/01/2023	IndReq		VSF			161-	0	
10/01/2023	PldOrd		000000425/STCK			142	142	0001
10/01/2023	IndReq		VSF			142-	0	
11/01/2023	PldOrd		000000426/STCK			141	141	0001
11/01/2023	IndReq		VSF			141-	0	
12/01/2023	PldOrd		000000427/STCK			162	162	0001
12/01/2023	IndReq		VSF			162-	0	
01/01/2024	PldOrd		000000428/STCK			177	177	0001
01/01/2024	IndReq		VSF			177-	0	

Choose  at the beginning of the **third** planned order. A popup opens with details about the order.

Additional Data for MRP Element

Plnd Order: 0000000425 Make-to-stock

Order End Date: 10/01/2023

GR pr.time: 0

Order Qty.: 142 EA

Order Start: 09/27/2023

Proc. Type: E

Scrap: 0

Opening Date: 09/26/2023

Order Type: LA

✓ ↺ ✎ ⚙ ⚙

[-> Prod.Ord.](#) [-> PartConvProdOrder](#) [-> Proc.Ord.](#) [-> SubProcOrd](#) [-> Pur Req.](#) ★ ✖

The planned order should now be converted into a production order. To do this, press [-> Prod.Ord.](#). The system creates a temporary production order, identified by the generic order number, and releases it automatically.

< SAP Production order Create: Header

Search In: "Apps"

Menu

Material Capacity Operations Components Documents Sequences

Order: %00000000001

Type: PP01

Material: DXTR3400 Deluxe Touring Bike (red)

Plant: DL00

Status: REL MACM SETC

General Assignment Goods Receipt Control Dates/Quantities Master Data Long Text Administration Items Fast Entry

Quantities

* Total Qty: 142 EA

Scrap Portion: 0.00 %

Delivered: 0

Short/Exc. Rcpt: 0

Note At this point, please note down the total quantity in your production order. You will need it later when confirming your order.

Total quantity

Determine the status of your order by clicking on [i](#).

Status Business processes

Syst. Status

X	Stat...	Text
☒	REL	Released
☒	MACM	Material committed
☒	SETC	Settlement rule created

Note When you converted the planned order to a production order, scheduling took place, an availability check was automatically carried out and a reservation was placed on the materials specified within the bill of materials.

Click on [<](#), to go back to the *Production order Create: Header* screen and save your production order with [Save](#).

Note When you save the production order the system will automatically calculate the planned costs for the production order.

The system assigns a unique number to the production order. Please make a note of the production order number.

 Σ 

Step 9: Receive Goods in Inventory

Task Receive goods in the Dallas plant.

Time 10 min

Short Description Receive enough goods in the Dallas storage locations to start the production process.

Name (Position) Susanne Castro (Goods Receipt Clerk)

Usually, at this point the purchasing department in Dallas would take over and procure enough raw materials from vendors to fill the inventory so that the production process can be initiated. In this case study, we are bypassing this procurement process (this process is explained in the MM unit in detail). Because the inventory for all DXTR3### components is empty, we will assume that we find 500 pieces each in the storage location.

Goods receipt

To receive goods in the inventory, in the space *Production Planning and Execution* space and the role of *Goods Receipt Clerk* use the app *Post Goods Receipt without Reference*.

Post Goods Receipt without Reference



The document and posting date are already defaulted with the current date and can thus be transferred.

Goods Receipt without Reference

General Information

Items

Attachments

Printing: No print

Note:

Document Date: 08/02/2023

Delivery Note:

Posting Date: 08/02/2023

Directly below you will find the *Item* section. The table there is ready for input and offers **item 01** in advance.

Items

Items							Copy	Delete	Create
☐ Item	Material	Quantity / Unit	Plant	Storage Location	Stock Type	Special Stocks			
☐ 01						None			

Click on the line of **item 01**, you will switch to a separate input window.

Enter your *Material* **TRWA1###** and press Enter.

TRWA1###

Now you can enter the *Quantity 500* with *Unit EA*.

Next, choose *Plant DL00*. When choosing the *Storage Location* your storage for semi-finished goods is will be proposed. You can check the status directly in the selection screen.

500
EA

DL00
SF00

Select Storage Location

Search

Standard

Semi-Fin. Goods

SF00

Unrestricted... 0.000 EA

Quality 0.000 EA

Blocked 0.000 EA

Choose the *Storage Location SF00*. The *Stock Type* is automatically set to **Unrestricted-Use**, and the *Special Stocks* is **None**.

Unrestricted-Use

None

Item 01

Material

Plant Options

Additional Information

Material: TRWA1400

Quantity / Unit: 500.000 EA

Plant Options

Plant: Plant Dallas (DL00)

Storage Location: Semi-Fin. Goods (SF00)

Stock Type: Unrestricted-Use

Special Stocks: None

Supplier:

Click on **Apply and New** to accept your entries and, at the same time, to be able to specify a new item. The system confirms the transfer of the item.

Item 01 is applied.

Now repeat the procedure for the other components of the bike DXTR3###.

Material	Quantity	Unit	Plant	SLoc
TRFR3### (Touring Frame-Red)	500	EA	DL00	RM00
DGAM1### (Derailleur Gear Assembly)	500	EA	DL00	RM00
TRSK1### (Touring Seat Kit)	500	EA	DL00	RM00
TRHB1### (Touring Handle Bar)	500	EA	DL00	RM00
PEDL1### (Pedal Assembly)	500	EA	DL00	RM00
CHAN1### (Chain)	500	EA	DL00	RM00
BRKT1### (Brake Kit)	500	EA	DL00	RM00
WDOC1### (Warranty Document)	500	EA	DL00	RM00
PCKG1### (Packaging)	500	EA	DL00	RM00

TRFR3###

DGAM1###

TRSK1###

TRHB1###

PEDL1###

CHAN1###

BRKT1###

WDOC1###

PCKG1###

As soon as you create the last item, confirm it with [Apply](#) to automatically return to the goods receipt posting. There you will see all created items.

☐	Item	Material	Quantity / Unit	Plant	Storage Location	Stock Type	Special Stocks
☐	01	TRWA1400	500.000 EA	Plant Dallas (DL00)	Semi-Fin. Goods (SFG0)	Unrestricted-Use	None
☐	02	TRFR3400	500.000 EA	Plant Dallas (DL00)	Raw Materials (RM00)	Unrestricted-Use	None
☐	03	DGAM1400	500.000 EA	Plant Dallas (DL00)	Raw Materials (RM00)	Unrestricted-Use	None
☐	04	TRSK1400	500.000 EA	Plant Dallas (DL00)	Raw Materials (RM00)	Unrestricted-Use	None
☐	05	TRHB1400	500.000 EA	Plant Dallas (DL00)	Raw Materials (RM00)	Unrestricted-Use	None
☐	06	PEDL1400	500.000 EA	Plant Dallas (DL00)	Raw Materials (RM00)	Unrestricted-Use	None
☐	07	CHAN1400	500.000 EA	Plant Dallas (DL00)	Raw Materials (RM00)	Unrestricted-Use	None
☐	08	BRKT1400	500.000 EA	Plant Dallas (DL00)	Raw Materials (RM00)	Unrestricted-Use	None
☐	09	WDOC1400	500.000 EA	Plant Dallas (DL00)	Raw Materials (RM00)	Unrestricted-Use	None
☐	10	PKHG1400	500.000 EA	Plant Dallas (DL00)	Raw Materials (RM00)	Unrestricted-Use	None

Note If you have forgotten an item, you can add further items by clicking on [Create](#). Furthermore, you can also correct entries if necessary.

Secure your goods receipt with [Post](#). The SAP system will assign a unique number to the goods receipt and issue an associated message.

Success

1 document(s) created:

Material Document 4900038023/2023

OK

Material Document
Number

Confirm the success message with [OK](#).

Click [SAP](#) to return to the SAP Fiori Launchpad.



Step 10: Issue Goods to Production Order

Task Issue goods to a production order.

Time 10 min

Short Description Now that all necessary components are on stock issue them to your production order in precise quantity.

Name (Position) Sanjay Datar (Warehouse Employee)

The goods issue process is fully defined in the production order, BOM, and routing. The quantities and the materials are reserved for this specific production order, they will be withdrawn with reference to the order number and will be used to assign actual costs to the production order for managerial accounting purposes.

To issue goods to a production order, in the space *Production Planning and Execution* space and the role of *Warehouse Employee* use the app *Post Goods Movement*.

Post Goods Movement



Make sure that **Goods Issue** and **Order** are selected in the drop-down menus.

A07 Goods Issue

R08 Order

The *Document Date* as well as the *Posting Date* should be already set to the current date. The *Movement Type* should be set to 261 (GI for order).

A07 Goods Issue

R08 Order

GI for order

261

General

Document Date: 08/02/2023

Posting Date: 08/02/2023

1 Individual Slip

Material Slip:

Doc. Header Text:

Enter your noted **production order** number.

Alternatively, click the value help icon  in the order field. In the *Order Number (1)* popup, use the icon on the far right  to display a list of all tabs. Select the tab *Production Orders using the Info System*. On this tab, enter your

Process Orders using
the Info System
DXTR3###

material **DXTR3###** in the *Material* field and click **Execute** . Select your order and accept it with .

Once you have found or entered your production order number, press to load the order details.

A07 Goods Issue

R08 Order

1000002

Note Goods issues posting for the required components is another milestone in the production order process. The following functions are performed when a GI for the components of the production order is posted:

- Storage-location-specific update of the stock and consumption fields
- Reduction of the reservation (for planned withdrawal)
- Update of costs for unplanned withdrawals
- Determination of actual costs (valuation) and order update
- Consumption update
- Generation of material and accounting documents (FI and CO documents)
- Creation of material document.
- Creation of accounting document
- Creation of controlling document
- Printing of GI document

The goods issues posting is controlled through a movement type (261), to which each posting refers. This can take place manually or automatically.

An itemized list will appear. It lists all the materials and their respective quantities that need to be issued to your order. You need to tell the system what Storage Location the materials should be withdrawn from. For the Touring Aluminum Wheel Assembly (TRWA1###), enter **SF00** (Semi-finished goods) and for all other materials **RM00** (Raw materials) in the SLoc fields. Before pressing Enter compare your screen with the one shown below. Notice that your quantity could be different.

SF00
RM00

Flag each item with **OK**. If you can't flag the first item please close the detail view with a click on under the line item list.

OK

Line	Mat. Short Text	Wa...	OK	Qty in UnE	EUn	S...	SLoc	Order	Acti...	Bus...	Co...
1	Touring Frame-Red	☐	☒	142	EA		RM00	1000002	0020	B100	US00
2	Deraillleur Gear Assembly	☐	☒	142	EA		RM00	1000002	0040	B100	US00
3	Touring Seat Kit	☐	☒	142	EA		RM00	1000002	0020	B100	US00
4	Pedal Assembly	☐	☒	142	EA		RM00	1000002	0070	B100	US00
5	Touring Handle Bar	☐	☒	142	EA		RM00	1000002	0030	B100	US00
6	Chain	☐	☒	142	EA		RM00	1000002	0050	B100	US00
7	Brake Kit	☐	☒	142	EA		RM00	1000002	0060	B100	US00
8	Warranty Document	☐	☒	142	EA		RM00	1000002	0100	B100	US00
9	Packaging	☐	☒	142	EA		RM00	1000002	0100	B100	US00
10	Touring Aluminum Wheel Assembly	☐	☒	284	EA		SF00	1000002	0040	B100	US00

Click on **Post** and record the material document number.

Material document 4900038024 posted

Material Document
Number

Click  to return to the SAP Fiori Launchpad.



Step 11: Review Production Order Status

Task Review the production order status.

Time 10 min

Short Description Review the current production order with respect to the status of the order.

Name (Position) Michael Brauer (Shop Floor Worker 4)

To display the production order, in the space *Production Planning and Execution* space and the role of *Shop Floor Worker* use the app *Manage Production Orders*.

Manage Production Orders - Orders



When you open the app for the first time, you will see a welcome message telling you to select an area of responsibility. Confirm this with **OK**, another popup will appear.

Area of Responsibility

Production Supervisor

>

Work Center

>

OK

Choose *Production Supervisor*. No Plant is currently assigned. Click on **+** an choose the **Plant Dallas**. Confirm your selection with **Select**.

Area of Responsibility (Supervisor)

Add Plant and Production Supervisor

+

Plant Dallas (DL00)

x

DL Production Scheduler (000)

OK

Cancel

Confirm your selection with **OK** now click on *Work Center*.

Click on  again and choose the **Work Center DL Assembly (ASSY1000)**, **DL Inspection (INSP1000)** and **DL Packaging (PACK1000)** from your Pland Dallas and click on .

Area of Responsibility (Work Center)

Add Plant and Work Center

Plant Dallas (DL00)

DL Assembly (ASSY1000)

Plant Dallas (DL00)

DL Inspection (INSP1000)

Plant Dallas (DL00)

DL Packaging (PACK1000)

OK

Cancel

Confirm your selection with , and press  to leave the selection or the area of Responsibility.

You will receive an overview of all existing orders. Depending on the progress of your course, there may be several production orders with different processing statuses.

Orders (3)

Order	Material	Open Quantity	Status	Start	End	Progress of Operation	Issues
☐ 1000001	DXTR3000 Deluxe Touring Bike (red)	0 EA	Delivered	Thu, Jul 13, 2023 06:55	Thu, Jul 13, 2023 07:55	Move to storage (0110) 141 of 141	  
☐ 1000000	DXTR3000 Deluxe Touring Bike (red)	0 EA	Delivered	Tue, Jul 11, 2023 05:30	Tue, Jul 11, 2023 06:30	Move to storage (0110) 142 of 142	  
☐ 1000002	DXTR3400 Deluxe Touring Bike (red)	142 EA	Released	Thu, Sep 21, 2023 09:05	Fri, Sep 29, 2023 17:00	Material staging (0010) 0 of 142	  

In the field *Material* enter your material **DXTR3###** and choose  to display your order.

DXTR3###

SAP Manage Production Orders

Search in "Apps"

Standard

Search

Status: Delivered x 6 More

Issue Type: No Filter x

Delay Duration: >= 0 Hours

Order:

Material: Deluxe Touring Bike x

Scheduled Start:

Go

Adapt Filters (5)

Orders (1)

Order	Material	Open Quantity	Status	Start	End	Progress of Operation	Issues
☐ 1000002	DXTR3400 Deluxe Touring Bike (red)	142 EA	Released	Thu, Sep 21, 2023 09:05	Fri, Sep 29, 2023 17:00	Material staging (0010) 0 of 142	  

The tabular overview already provides you with various information about your order, such as the current status and the current processing status.

For further information select the entry. You will be forwarded to the details of the production order.

< **SAP** Manage Production Orders

Search In: "Apps"

1000002

Edit OrderReleaseRead Master Data

Material:DXTR3400 (Deluxe Touring Bike (red))

Status:Released

Person Responsible:DL MRP Controller

Quantity:142.000 EA

Issues

Order Information

Components

Order Schedule

Confirmation

Inspection

Order Information

General

Dates/Times

Quantities

Production Plant:
DL00

MRP Area:
DL00 Plant Dallas

Planning Plant:
DL00 Plant Dallas

Production Version:
0001 Generated Version 0001

MRP Controller:
000

Scheduled Start:
Thu, Sep 21, 2023, 09:05

Scheduled End:
Fri, Sep 29, 2023, 17:00

Planned Start :
Wed, Sep 20, 2023, 00:00

Planned End:
Sun, Oct 1, 2023, 00:00

Scheduled Release:
Wed, Sep 20, 2023

Total Quantity:
142 EA

Confirmed Yield:
0 EA

Confirmed Scrap:
0 EA

GR Quantity:
0 EA

Open Quantity:
142 EA

Click on the status **Released** for more information. You can see that your production order is precosted and a settlement rule has been created.

1000002

Material:
DXTR3400 (Deluxe Touring Bike (red))

Status:
Released

Status Overview

Released

Precosted

Settlement Rule Created

Now click on the *Components* tab. The screen scrolls to the corresponding position.

Components

AllMissing

Material

Quantity

Coverage

Backflush

Component Scrap

Requirement Date/Time

Storage Location

TRFR3400 Touring Frame-Red	Total Quantity: 142 EA Open Quantity: 0 EA	142 EA	No	0.00 %	Thu, Sep 21, 2023 10:51	
TRSK1400 Touring Seat Kit	Total Quantity: 142 EA Open Quantity: 0 EA	142 EA	No	0.00 %	Thu, Sep 21, 2023 10:51	
TRHB1400 Touring Handle Bar	Total Quantity: 142 EA Open Quantity: 0 EA	142 EA	No	0.00 %	Thu, Sep 21, 2023 13:31	
DGAM1400 Derailleur Gear Assembly	Total Quantity: 142 EA Open Quantity: 0 EA	142 EA	No	0.00 %	Fri, Sep 22, 2023 09:51	

In the last task, you posted the goods issue for the production order. In the production order, you now see that there are no more open quantities for this order.

Click **SAP** to return to the SAP Fiori Launchpad.

Step 12: Confirm Production Completion

Task Confirm production order completion.

Time 10 min

Short Description Confirm completion for your production order.

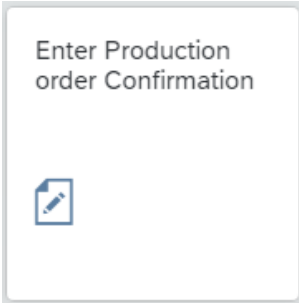
Name (Position) Michael Brauer (Shop Floor Worker 4)

When the assembly has been completed for the current production order, we need to confirm that certain procedures and activities have been completed and record the quantity of the finished product that has been manufactured.

Production completion

To confirm production completion, in the space *Production Planning and Execution* space and the role of *Shop Floor Worker* use the app *Enter Production order Confirmation*.

Enter Production order Confirmation



Enter your **production order** number and click on **Enter**.

Production order number

Alternatively, use the Input-Help-Symbol  and switch in the popup with  to the tab *production orders by material and routing*, enter your material number **DXTR3##** there and search for your order.

Check if **Final Confirmation** and **Clear Open Reserv.** is already checked in the *Confirmation Type* section.

Final Confirm.
Clear Reservation
Amount

< **SAP** Confirmation of Production Order Enter : Actual Data

Menu 

Order: 1000002

Status: REL PRC GMPS MACM SETC

Material: DXTR3400

Material Descr.: Deluxe Touring Bike (red)

Confirmation Type

Partial confirmation: ☐

Clear Open Reservs.: ☒

Final Confirmation: ☒

Autom. Final Conf.: ☐

Furthermore, in the *Actual Data* tab, the quantity of bicycles that you should produce for this order should already be entered in the *Yield Quantity* field. Change the *Start Execution* to **1 hour earlier** than the preset time.

1 hour earlier

Actual Data

	Curr. t/b Conf.	Unit	Confirmed to Date	Planned t/b Conf.	Unit
Yield Quantity:	142	EA	0	142	EA
Scrap Quantity:			0	0	
Rework Quantity:			0		
Reason for Var.:					
Personnel no.:					

	To Be Confirmed	Confirmed to Date	Planned t/b Conf.
Start Execution:	08/02/2023 07:02:00		09/21/2023
Finish Execut.:	08/02/2023 08:02:00		09/29/2023
Posting Date:	08/02/2023		

Confirm. Text:

Long Text exists: ☐

Save your entries with . You will get a confirmation from the system.

 Confirmation of order 1000002 saved

Note When the confirmation is saved, labor costs for the order are calculated automatically. The quantity yield also establishes the parameters for the goods receipt into Inventory.

Click  to return to the SAP Fiori Launchpad.



Step 13: Review Production Order Status

Task Review the production order status.

Zeit 10 Min.

Short Description Review the current production order with respect to the status of the order.

Name (Position) Michael Brauer (Shop Floor Worker 4)

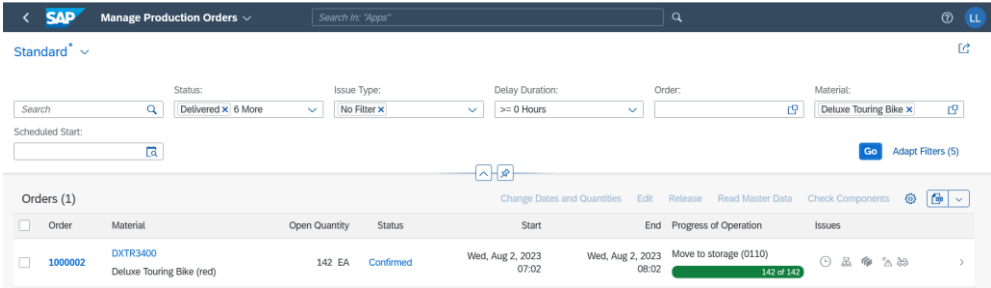
To display the production order, in the space *Production Planning and Execution* space and the role of *Shop Floor Worker* use the app *Manage Production Orders*.

Manage Production Orders



In the field *Material* enter your material **DXTR3###** and choose **Go** to display your order.

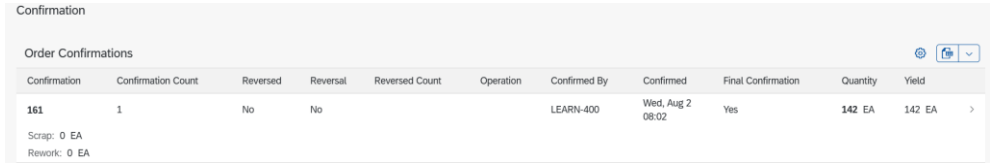
DXTR3###



As you can see, the status of your production order has changed from *Released* to *Confirmed*. Furthermore, the processing status is now at *Move to storage*.

For more information, select the entry, you will be redirected to the details of the production order. Click on the *Confirmation* tab to go to the related area.

Confirmation



An order confirmation is now available. You can see that the complete quantity of your production order has been confirmed and that there is no scrap.

After the confirmation, the goods receipt now needs to take place to complete the order.

Click  to return to the SAP Fiori Launchpad.



Step 14: Receive Goods from Production Order

Task Post a goods receipt from production order.

Time 15 min

Short Description Post a goods receipt from your production order.

Name (Position) Susanne Castro (Goods Receipt Clerk)

Receive the completed products into finished goods inventory. Check the quantity proposed against the quantity specified in the production order and the quantity specified during confirmation. If there are any discrepancies, the system will decide if an error or warning message should be generated, depending upon the deviation identified.

Goods receipt

To post the goods receipt, in the space *Production Planning and Execution* space and the role of *Goods Receipt Clerk* use the app *Post goods receipt for production order*.

Post goods receipt for production order



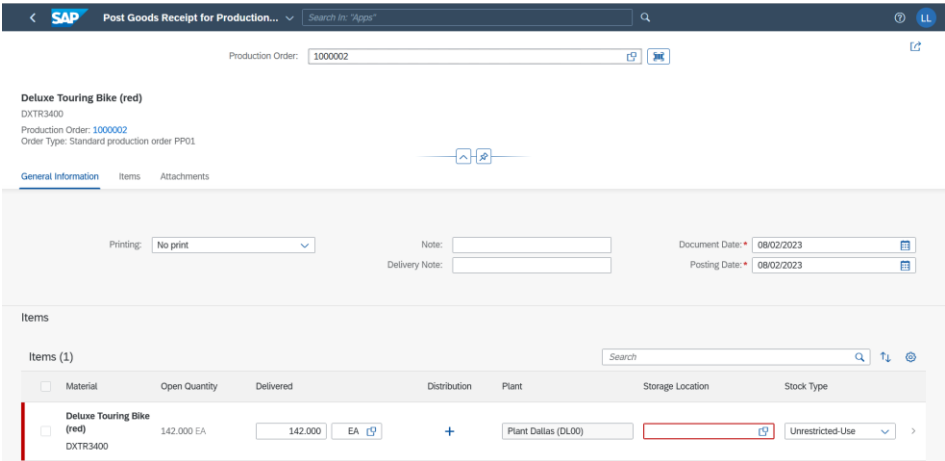
Enter your noted **production order** and press Enter.

Production order

As an alternative use the Input-Help-Search  and enter your material **DXTR3###** and click . Then select your production order from the result list.

DXTR3###

Your production order is loaded and displayed.



Add the storage location **FG00** for end products, all other settings can be accepted.

Items

Items (1)

Search

Q

↑↓

⚙

☒	Material	Open Quantity	Delivered	Distribution	Plant	Storage Location	Stock Type
☒	Deluxe Touring Bike (red) DXTR3400	142.000 EA	142.000	EA	+	Plant Dallas (DL00)	Finished Goods (FG00)
							Unrestricted-Use

Save your goods receipt with **Post**. The SAP system will assign a unique number to the goods receipt and issue a corresponding message.

✔ Success

1 document(s) created:

🕒 Material Document 5000000017/2023

OK

Material Document
Number

Furthermore, this updates the current value of the material produced to the production order.

Confirm the successmessage with **OK**.

Click **SAP** to return to the SAP Fiori Launchpad.



Step 15: Review Costs Assigned to Production Order

Task View the corresponding costs assigned to your production order.

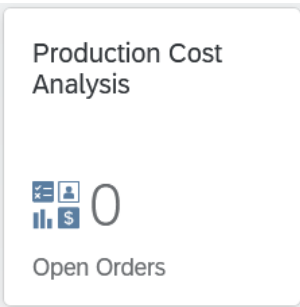
Description Review all costs associated with your production order.

Name (Position) Jamie Shamblin (Controller)

Zeit 5 Min.

To view the cost of a manufacturing job, in the space *Production Planning and Execution* space and the role of *Controller* use the *Production Cost Analysis* app.

Production Cost Analysis



In the *Product* field enter **DXTR3###** and change the *Order Status* from **Open** to **Closed**

DXTR3###
Closed

SAP

Production Cost Analysis

Search In: "Apps"

Standard

Profit Center:

Plant:

Order Type:

Product:

Order:

Order Status: *

Period From: *

Period To: *

008/2023

008/2023

Go

Clear

Adapt Filters (5)

Click on **Go** to start the search. Your recently completed production order will be displayed.

Order Analysis

Cost Component Group

Order List (1)

Standard

Orders

Exception Status	Order	Product	Tgt DR/Actl DR Varc	Target Cost Debit	Actl Cost Debit	Total Actl Cost	Order Status	Op...
①	1000002	DXTR3400 (Deluxe To...	103,732.55 USD	0.00 USD	103,732.55 USD	-218.55 USD	Closed	>
			103,732.55 USD	0.00 USD	103,732.55 USD	-218.55 USD		

This overview lists the summed target and actual costs and shows any variances. Click on **>** at the end of the entry to open the cost details.

COST DETAILS

Standard

G/L Account	Total Target Cost	Total Actl Cost	Tgt/Actl CostVar	Actual Quantity	Origin	Origin Activity Type Name
Group Name:						
7520000 (Settle. Manuf Ou...	0.00 USD	-103,951.10 USD	-103,951.10 USD	-142 EA	DL00DXTR3400	
Group Name: Human Resources						
8000000 (Labor)	0.00 USD	3,551.55 USD	3,551.55 USD	71.031 HR	NAPR1000LABOR	Labor hours
Group Name: Raw Materials						
5001000 (RM Consumpt E...	0.00 USD	9,940.00 USD	9,940.00 USD	142 EA	DL00BRKT1400	
5001000 (RM Consumpt E...	0.00 USD	1,420.00 USD	1,420.00 USD	142 EA	DL00CHAN1400	
5001000 (RM Consumpt E...	0.00 USD	10,650.00 USD	10,650.00 USD	142 EA	DL00DGAM1400	
5001000 (RM Consumpt E...	0.00 USD	497.00 USD	497.00 USD	142 EA	DL00PCGL1400	
5001000 (RM Consumpt E...	0.00 USD	6,390.00 USD	6,390.00 USD	142 EA	DL00PEDL1400	
5001000 (RM Consumpt E...	0.00 USD	28,400.00 USD	28,400.00 USD	142 EA	DL00TRFR3400	
5001000 (RM Consumpt E...	0.00 USD	3,550.00 USD	3,550.00 USD	142 EA	DL00TRHB1400	
5001000 (RM Consumpt E...	0.00 USD	7,100.00 USD	7,100.00 USD	142 EA	DL00TRSK1400	
5001000 (RM Consumpt E...	0.00 USD	142.00 USD	142.00 USD	142 EA	DL00WDOC1400	
	0.00 USD	-218.55 USD	-218.55 USD	1,491.031 *		

Now that the finished products have been received in the Inventory, the Manufacturing Output Settlement Variance has been added. How is this figure calculated by the system?

Click  to return to the SAP Fiori Launchpad.



Step 16: Settle Costs of Production Order

Task Settle costs of your production order.

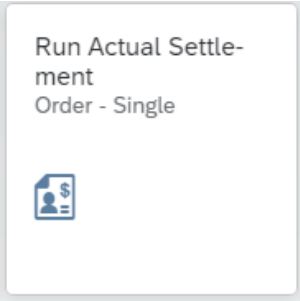
Time 20 min

Short Description Settle the costs of your production order. The costs are temporarily captured in the production order and they need to be assigned to an appropriate cost object. Compare the actual costs to the planned costs to identify any deviations or potential problems in this regard.

Name (Position) Jamie Shamblin (Controller)

To settle costs of a production order, in the space *Production Planning and Execution* space and the role of use the app *Run Actual Settlement - Order - Single*.

Run Actual Settlement -
Order - Single



If you have to input the Controlling Area, enter **NA00**, and click on **Continue**.

NA00

Enter your **production order number**, alternatively search for it with your material **DXTR3###** in the Input-Help. In the *Parameters* section enter *Settlement- and posting period* as **current month** (e.g., 006 for June). Enter as *Fiscal Year* the **current year**. Make Sure that **Test Run** is checked in the *Processing Options* section.

Orderer number

current month
current year

Test Run

< SAP

Actual Settlement: Order

Search In: "Apps"

Menu

Settlement Rule

Controlling Area: NA00

* Order: 1000002

Parameters

* Settlement Period: 008

* Fiscal Year: 2023

* Processing Type: 1 Automatic

Posting Period: 008

Asset Value Date:

Processing Options

☒ Test Run

☐ Check Trans. Data

Click on **Execute** to proceed. Confirm any occurring pop-ups with enter. You enter the screen *Actual Settlement: Oder Basic list*.

< **SAP**

Actual Settlement: Order Basic list

Search In: "Apps"

Menu ▾

Selection

Selection Parameters	Value	Name
Order	1000002	Deluxe Touring Bike (red)
Period	008	
Posting Period	008	
Fiscal Year	2023	
Processing Type	1	Automatic
Posting Date	08/31/2023	
Controlling Area	NA00	Global Bike North America
Currency	USD	United States Dollar
Value Date	08/31/2023	

Processing Options

Selection Parameters	Value
Execution Type	Settlement Executed
Processing Mode	Test run

Processing completed with no errors

Statistics

Processing Category	⌘	Number
Settlement Executed		1
No Change		
Not Relevant		
Inappropriate Status		
Error		

Click on  to open the detail lists.

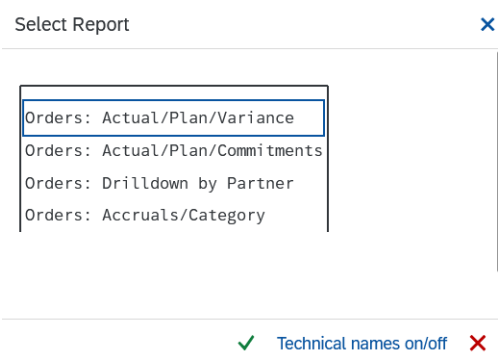
Detail list - Settled values					
Senders	Short text: Sender	Receiver	⌘	Val/COArea Crcy	Additional information
ORD 1000002	Deluxe Touring Bike (red)	MAT DL00/DXTR3400		218.55-	
			•	218.55-	

From the Menu bar choose:

Menu ► Environment ► Report

The *Select Report* pop-up will appear.

Menu Bar



Click on **Actual/Plan/Variance** and confirm your selection with . A corresponding report group is generated and displayed.

Actual/Plan/Variance

Orders: Actual/Plan/Variance

Date: 08/02/2023 15:26:02

Page: 2 / 2

Order/Group

1000002

000001000002

Fiscal year

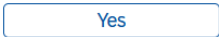
2023

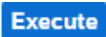
Period

1 - 8

Cost Elements	Actual	Plan	Var. (Abs.)	Var. (%)
5001000 RM Consumpt Expense	68,089.00		68,089.00	
5004000 SF Consumpt Expense	32,092.00		32,092.00	
8000000 Labor	3,551.55		3,551.55	
8015000 Material Overhead				
* Costs	103,732.55		103,732.55	
7520000 Settle. Manuf Output	103,951.10-		103,951.10-	
* Deliveries to Stock	103,951.10-		103,951.10-	
** Balance	218.55-		218.55-	

The test run is now complete and the *actual settlement* should now be performed.

Click on  to go back. Confirm the Message to leave the report with . Afterwards press  two times to get back to the entry screen.

Deselect Test Run and again click on . In contrast to the previous run, you can now see in the *Processing Options* area that this was an update run that was *completed with no errors*.

Testlauf

Processing Options

Selection Parameters	Value
Execution Type	Settlement Executed
Processing Mode	Update run
Processing completed with no errors	

Open the report **Actual/Plan/Variance** again by clicking on  first and then use the menu bar path **Menu ▶ Environment ▶ Report** and select the option **Actual/Plan/Variance**.

Actual/Plan/Variance

Orders: Actual/Plan/Variance		Date: 08/02/2023 15:29:16	Page: 2 / 2
Order/Group	1000002	000001000002	
Fiscal year	2023		
Period	1 - 8		

Cost Elements	Actual	Plan	Var. (Abs.)	Var. (%)
5001000 RM Consumpt Expense	68,089.00		68,089.00	
5004000 SF Consumpt Expense	32,092.00		32,092.00	
8000000 Labor	3,551.55		3,551.55	
8015000 Material Overhead				
* Costs	103,732.55		103,732.55	
7520000 Settle. Manuf Output	218.55		218.55	
* Settled Costs	218.55		218.55	
7520000 Settle. Manuf Output	103,951.10-		103,951.10-	
* Deliveries to Stock	103,951.10-		103,951.10-	
** Balance				

You can see that the costs have now been settled.

Click  to return to the SAP Fiori Launchpad.



PP Challenge

Learning Objective Understand and perform an integrated manufacturing process.

Time 60 min

Motivation After you have successfully worked through the *Production Planning and Execution* case study you should be able to solve the following challenge on your own.

Scenario In this challenge you should create sales and operations plan (SOP) for the product group (product family) Mountain bikes. Take into consideration that the materials of the product group have to be assigned to the strategy group. Therefore, enter manually the following sales figures:

Period	Sales (volume)
Current month + 2	150
Current month + 3	175
Current month + 4	200
Current month + 5	85
Current month + 6	90
Current month + 7	115

In addition, you must post the correct goods for Material ORMN1### in the storage location in order to be able to produce and settle costs afterwards.

Task Information Since this task is based on the *Production Planning and Execution* case study you can use it as guidance. However, it is recommended that you solve it without any help in order to test your acquired knowledge.

